Depropanizer Column Ver 1 · OTS

DC3-V1-PC-PKG-001

Procurement package list

Rev 0 · Issued for teaching

Unit	Depropanizer Column Ver 1 · OTS
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Note. *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

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1 · Purpose and the EPC phases after engineering

Purchase packages of the C-201 depropanizer unit, generated from the equipment list DC3-V1-ME-EQL-001, the instrument list DC3-V1-IN-INL-001, the line list DC3-V1-PE-LIN-001 and the electrical load list DC3-V1-EE-LL-001. Each package carries its items, the documents that specify it (sent with the enquiry), the assumed lead time and its class, the inspection level and the FAT, hold and witness points, and a critical-path flag from the planned order dates. Procurement issues the enquiries, the discipline engineers evaluate the bids technically, the project manager approves the purchase orders; expediting and inspection follow the vendor to delivery.

Phase	Weeks (assumed)	Dates	Note
Detailed engineering	0–36	04-Jan-27 – 13-Sep-27	from the FEED package to IFC drawings
Procurement	2–56	18-Jan-27 – 31-Jan-28	enquiry, bid evaluation, PO, expediting, delivery
Fabrication, inspection and FAT	8–56	01-Mar-27 – 31-Jan-28	shop work, witness and hold points, FAT
Construction	26–78	05-Jul-27 – 03-Jul-28	civil, erection, piping, electrical, instruments
Mechanical completion	70–78	08-May-28 – 03-Jul-28	system by system; punch A cleared
Pre-commissioning	66–82	10-Apr-28 – 31-Jul-28	flushing, tests, drying, N_2, loop checks
Commissioning	80–86	17-Jul-28 – 28-Aug-28	utilities in service, gassing-up, ready for start-up
Start-up	86–87	28-Aug-28 – 04-Sep-28	first feed to line-out (the OTS procedure)
Performance test	88–89	11-Sep-28 – 18-Sep-28	72 h at the guarantee conditions
Handover	90	25-Sep-28	provisional acceptance, care and custody to the owner

2 · Package summary and planned schedule

Package	Items	Lead weeks	Class	Inspection	PO (week)	On site	Needed (MC − offset)	Float weeks	Critical
PKG-01 Column C-201 with trays and internals	1	44	L1	Level 1	15-Feb-27 (wk 6)	20-Dec-27 (wk 50)	03-Jan-28 (MC − 26)	2	✗ critical
PKG-02 Drums V-201 and V-202	2	32	L2	Level 2	15-Mar-27 (wk 10)	25-Oct-27 (wk 42)	20-Dec-27 (MC − 28)	8	—
PKG-03 Shell-and-tube exchangers E-201, E-202, E-203	3	36	L2	Level 2	01-Mar-27 (wk 8)	08-Nov-27 (wk 44)	17-Jan-28 (MC − 24)	10	—
PKG-04 Pumps P-201A/B, P-202A/B with motors and the P-201B steam turbine	8	48	L1	Level 1	01-Mar-27 (wk 8)	31-Jan-28 (wk 56)	14-Feb-28 (MC − 20)	2	✗ critical
PKG-05 Control valves	7	26	L2	Level 2	26-Apr-27 (wk 16)	25-Oct-27 (wk 42)	27-Mar-28 (MC − 14)	22	—
PKG-06 SIS on/off valves XV-201, XV-202, XV-209	6	30	L2	Level 2	12-Apr-27 (wk 14)	08-Nov-27 (wk 44)	27-Mar-28 (MC − 14)	20	—
PKG-07 Pressure safety valve PSV-201	1	20	L3	Level 2	10-May-27 (wk 18)	27-Sep-27 (wk 38)	24-Apr-28 (MC − 10)	30	—
PKG-08 Field instruments	41	18	L3	Level 3	24-May-27 (wk 20)	27-Sep-27 (wk 38)	10-Apr-28 (MC − 12)	28	—
PKG-09 Process analyzers AT-219, AT-220 with shelter	3	36	L2	Level 2	12-Apr-27 (wk 14)	20-Dec-27 (wk 50)	10-Apr-28 (MC − 12)	16	—
PKG-10 Piping bulks, hand valves and steam traps	20	14	L4	Level 4	21-Jun-27 (wk 24)	27-Sep-27 (wk 38)	06-Dec-27 (MC − 30)	10	—
PKG-11 Electrical: MCC, UPS, lighting, space heaters, cables	6	28	L2	Level 3	10-May-27 (wk 18)	22-Nov-27 (wk 46)	14-Feb-28 (MC − 20)	12	—
PKG-12 DCS and SIS (control system)	52	32	L2	Level 1	29-Mar-27 (wk 12)	08-Nov-27 (wk 44)	13-Mar-28 (MC − 16)	18	—

With the planned order dates every package is on site by its need date; PKG-01, PKG-04 have 4 weeks of float or less (query EC-05). The page's schedule explorer moves the order dates, the lead-time factor and the MC target.

3 · Package sheets

PKG-01 · Column C-201 with trays and internals

Items	**C-201** Depropanizer column — 1.7 m ID × 23.0 m T/T; tray spacing 0.61 m; feed tray 18 27 sieve trays with supports and downcomers (CS, assumed) Skirt, base ring and anchor-bolt template Platform and ladder clips (shop-welded)
Specification documents	DC3-V1-ME-EQL-001, DC3-V1-ME-DS-001, DC3-V1-PE-HMB-001, DC3-V1-PID-001
Lead time (assumed)	44 weeks · class L1 · PO week 6 → on site week 50 · needed week 52 · float 2 weeks · CRITICAL
Inspection	Level 1 — full: pre-inspection meeting, hold points on materials, welding, NDT and the pressure test, final inspection, document review
FAT / hold / witness points	**H** Plate certificates EN 10204 3.1 and impact tests at −46 °C **R** Welding procedures and welder qualifications **W** NDT (spot radiography, E 0.85) and PWHT records **H** Shop hydrotest (horizontal) at the UG-99 pressure **W** Tray levelness, weir height and tray-ring position **W** Final dimensions, nozzle orientation, painting; release note

PKG-02 · Drums V-201 and V-202

Items	**V-201** Feed surge drum — 2.1 m ID × 6.5 m T/T (24.8 m³ from the model, L/D ≈ 3 assumed) **V-202** Reflux drum — 1.8 m ID × 5.6 m T/T (15.7 m³ from the model) Saddles, level-gauge and instrument nozzles
Specification documents	DC3-V1-ME-EQL-001, DC3-V1-ME-DS-001, DC3-V1-PE-HMB-001, DC3-V1-PID-001
Lead time (assumed)	32 weeks · class L2 · PO week 10 → on site week 42 · needed week 50 · float 8 weeks
Inspection	Level 2 — witness: the pressure test / performance test / FAT witnessed, final inspection, document review
FAT / hold / witness points	**R** Plate certificates and impact tests at −46 °C **R** Welding procedures, NDT records **W** Shop hydrotest at the UG-99 pressure **W** Final inspection and release note

PKG-03 · Shell-and-tube exchangers E-201, E-202, E-203

Items	**E-201** Feed preheater — 90 m² (= model UA 45 kW/K ÷ U 500 W/m² K assumed); duty 3.50 MW **E-202** Overhead condenser — 290 m² (= model clean UA 203 kW/K ÷ U 700 W/m² K assumed); duty 3.96 MW **E-203** Reboiler — 73 m² (= model UA 73 kW/K ÷ U 1000 W/m² K assumed); duty 2.24 MW Tube bundles, spare gaskets (one set, assumed)
Specification documents	DC3-V1-ME-EQL-001, DC3-V1-ME-DS-001, DC3-V1-PE-HMB-001, DC3-V1-PID-001
Lead time (assumed)	36 weeks · class L2 · PO week 8 → on site week 44 · needed week 54 · float 10 weeks
Inspection	Level 2 — witness: the pressure test / performance test / FAT witnessed, final inspection, document review
FAT / hold / witness points	**R** Thermal and mechanical design (TEMA R) against the datasheet **W** Tube-to-tubesheet joint mock-up and leak test **W** Hydrotest of each side separately (shell, then tubes) **W** Final inspection, bundle pull check (E-202 AES)

PKG-04 · Pumps P-201A/B, P-202A/B with motors and the P-201B steam turbine

Items	**P-201A/B** Feed pumps — Rated 77 m³/h × 1.71 MPa (317 m); absorbed 56.2 kW at rated; motor 75 kW **P-202A/B** Reflux / product pumps — Rated 102.5 m³/h × 0.462 MPa (100 m); absorbed 22 kW at rated; motor 30 kW P-201A — Feed pump (75 kW) P-201B — Feed pump (spare), steam-turbine drive (non-electric) P-202A — Reflux / product pump (30 kW) P-202B — Reflux / product pump (spare) (30 kW) Mechanical seals with seal-flush plans Couplings, baseplates; turbine governor and trip-and-throttle valve
Specification documents	DC3-V1-ME-EQL-001, DC3-V1-ME-DS-001, DC3-V1-PE-HMB-001, DC3-V1-EE-LL-001, DC3-V1-PID-001
Lead time (assumed)	48 weeks · class L1 · PO week 8 → on site week 56 · needed week 58 · float 2 weeks · CRITICAL
Inspection	Level 1 — full: pre-inspection meeting, hold points on materials, welding, NDT and the pressure test, final inspection, document review
FAT / hold / witness points	**H** Casing hydrotest (API 610: 1.5 × casing MAWP) **H** Performance test: head, flow, efficiency, NPSHr at the rated point **W** Mechanical run test (vibration, bearing temperature) **W** Turbine solo run and overspeed trip test (P-201B) **R** Motor routine test certificates

PKG-05 · Control valves

Items	7 instrument-list rows: FV-201, TV-202, FV-204, TV-208, LV-202, LV-203, PV-210 Positioners, air sets, limit switches as in the INL
Specification documents	DC3-V1-IN-INL-001, DC3-V1-PE-HMB-001, DC3-V1-PE-LIN-001, DC3-V1-PID-001
Lead time (assumed)	26 weeks · class L2 · PO week 16 → on site week 42 · needed week 64 · float 22 weeks
Inspection	Level 2 — witness: the pressure test / performance test / FAT witnessed, final inspection, document review
FAT / hold / witness points	**W** Body hydrotest and seat-leakage test (IEC 60534-4) **W** Stroke test and fail position on loss of air **R** Cv sizing report against the HMB cases

PKG-06 · SIS on/off valves XV-201, XV-202, XV-209

Items	6 instrument-list rows: XV-201, ZSC-201, XV-202, ZSC-202, XV-209, ZSC-209 Spring-return actuators, solenoids, limit switches
Specification documents	DC3-V1-IN-INL-001, DC3-V1-IN-CE-001, DC3-V1-IN-SRS-001, DC3-V1-PID-001
Lead time (assumed)	30 weeks · class L2 · PO week 14 → on site week 44 · needed week 64 · float 20 weeks
Inspection	Level 2 — witness: the pressure test / performance test / FAT witnessed, final inspection, document review
FAT / hold / witness points	**W** Seat-leakage test (tight shut-off) **H** Stroke time on de-energising the solenoid (≤ 2 s, SRS assumed) **R** SIL certificate / failure-rate data of valve, actuator and solenoid

PKG-07 · Pressure safety valve PSV-201

Items	1 instrument-list rows: PSV-201 Inlet / outlet isolation valves (locked open), spare (assumed)
Specification documents	DC3-V1-PE-RLB-001, DC3-V1-IN-INL-001, DC3-V1-ME-EQL-001, DC3-V1-PID-001
Lead time (assumed)	20 weeks · class L3 · PO week 18 → on site week 38 · needed week 68 · float 30 weeks
Inspection	Level 2 — witness: the pressure test / performance test / FAT witnessed, final inspection, document review
FAT / hold / witness points	**W** Set-pressure (cold differential test pressure) test **W** Seat tightness (API 527) at 90 % of set **R** Orifice and capacity certificate against the relief load

PKG-08 · Field instruments

Items	41 instrument-list rows: FE-201, FT-201, TT-202, FE-204, FT-204, TT-227, TT-208, LT-202, LT-203, PT-210, LT-201, TT-201, FT-206, PT-207, TT-241, TT-242, TT-243, TT-245, TT-246, TT-209, PT-211, PDT-212, FT-205, PT-205, FT-202, FT-203, TT-213, FT-216, TT-214, TT-215, FT-217, FE-202, FE-203, FE-205, FE-206, FE-216, FE-217, LS-202, LS-203, PDT-230, LG-201 Thermowells, manifolds, impulse tubing
Specification documents	DC3-V1-IN-INL-001, DC3-V1-PE-LIN-001, DC3-V1-PID-001
Lead time (assumed)	18 weeks · class L3 · PO week 20 → on site week 38 · needed week 66 · float 28 weeks
Inspection	Level 3 — final: final visual and dimensional inspection and certificate review before release for shipment
FAT / hold / witness points	**R** Calibration certificates (range as the INL) **R** Orifice-plate bore calculations against the line list **W** Hazardous-area certificates (Ex) against the area classification

PKG-09 · Process analyzers AT-219, AT-220 with shelter

Items	AH-219/220 — Analyzer house: AI-219 / AI-220 gas chromatographs, sample conditioning, HVAC and purge (10 kW) 2 instrument-list rows: AT-220, AT-219 Sample conditioning systems, calibration gas cylinders
Specification documents	DC3-V1-IN-INL-001, DC3-V1-PE-HMB-001, DC3-V1-EE-LL-001, DC3-V1-PID-001
Lead time (assumed)	36 weeks · class L2 · PO week 14 → on site week 50 · needed week 66 · float 16 weeks
Inspection	Level 2 — witness: the pressure test / performance test / FAT witnessed, final inspection, document review
FAT / hold / witness points	**W** Analyzer house FAT: calibration with certified gas at the spec level (2.5 / 2.0 mol %) **W** Response time from the sample tap to the DCS **R** HVAC and gas detection of the shelter

PKG-10 · Piping bulks, hand valves and steam traps

Items	20 lines: 6"-P-200 (A1L), 6"-P-201 (A1L), 4"-P-202 (B1L), 6"-P-203 (B1L), 8"-P-204 (B1L), 6"-P-205 (B1L), 1"-P-206 (B1L), 6"-P-207 (B1L), 4"-P-208 (B1L), 4"-P-209 (B1L), 4"-P-210 (B1L), 8"-P-211 (B1L), 10"-P-212 (B1L), 6"-LS-301 (A1S), 6"-LS-302 (A1S), 10"-CW-401 (A1W), 10"-CW-402 (A1W), 3"-P-213 (B1L), 2"-CD-501 (A1C), 2"-CD-502 (A1C) Pipe, fittings, flanges, gaskets, bolting by piping class (MTO from the 3D model) Valve list items (moved out of the INL): ST-201, ST-202, HV-202, HV-209, HV-210, HV-216
Specification documents	DC3-V1-PE-LIN-001, DC3-V1-PID-001, DC3-V1-ME-EQL-001
Lead time (assumed)	14 weeks · class L4 · PO week 24 → on site week 38 · needed week 48 · float 10 weeks
Inspection	Level 4 — documents: certificates (EN 10204 3.1) reviewed on receipt at site
FAT / hold / witness points	**R** Material certificates and positive material identification (alloy and LT carbon steel) **R** Valve shell and seat test certificates (API 598)

PKG-11 · Electrical: MCC, UPS, lighting, space heaters, cables

Items	UPS-DCS — DCS controllers, I/O, operator console share, P-201B turbine solenoid and governor (5 kVA) UPS-SIS — SIS logic solver, SIS I/O, solenoids of XV-201 / XV-202 / XV-209 (2 kVA) LTG-200 — Area and platform lighting (LED, 10 W/m², assumed; 10 % of fittings with own battery) (6 kW) SP-200 — Small power: 2 × 63 A welding / maintenance outlets, convenience sockets (20 kW) SH-200 — Motor anti-condensation heaters (3 × 0.1 kW, on while the motor is stopped) (0.3 kW) HT-200 — Heat tracing (non-electric) MCC buckets for P-201A, P-202A, P-202B Cables sized in the Electrical tab
Specification documents	DC3-V1-EE-LL-001, DC3-V1-PID-001
Lead time (assumed)	28 weeks · class L2 · PO week 18 → on site week 46 · needed week 58 · float 12 weeks
Inspection	Level 3 — final: final visual and dimensional inspection and certificate review before release for shipment
FAT / hold / witness points	**W** MCC routine tests (IEC 61439) **W** UPS autonomy and change-over test **R** Cable test certificates

PKG-12 · DCS and SIS (control system)

Items	52 instrument-list rows: FIC-201, TIC-202, FIC-204, TIC-227, TIC-208, LIC-202, LIC-203, PIC-210, LI-201, LIC-201, TI-201, FI-206, PI-207, TI-241, TI-242, TI-243, TI-245, TI-246, TI-209, PI-211, PDI-212, FI-205, PI-205, FI-202, AI-220, FI-203, AI-219, TI-213, FI-216, TI-214, TI-215, FI-217, PDI-230, LSLL-202, LSHH-203, HS-ESD, HS-RST, FSLL-201, PDSL-230, HY-201A, YI-201A, HY-201B, YI-201B, HY-202A, YI-202A, HY-202B, YI-202B, XA-201A, XA-202A, XA-202B, HS-201B, HS-202B Operator console, engineering station, cabinets, marshalling Configuration of the controllers, alarms and the C&E logic
Specification documents	DC3-V1-IN-INL-001, DC3-V1-IN-CE-001, DC3-V1-IN-SRS-001, DC3-V1-PID-001
Lead time (assumed)	32 weeks · class L2 · PO week 12 → on site week 44 · needed week 62 · float 18 weeks
Inspection	Level 1 — full: pre-inspection meeting, hold points on materials, welding, NDT and the pressure test, final inspection, document review
FAT / hold / witness points	**H** FAT with simulated I/O: every loop, alarm and C&E row (part 4) **W** Redundancy and diagnostics tests **R** SIS safety manual and SIL verification

4 · Inspection levels and lead-time classes

Level	Scope
1	Level 1 — full: pre-inspection meeting, hold points on materials, welding, NDT and the pressure test, final inspection, document review
2	Level 2 — witness: the pressure test / performance test / FAT witnessed, final inspection, document review
3	Level 3 — final: final visual and dimensional inspection and certificate review before release for shipment
4	Level 4 — documents: certificates (EN 10204 3.1) reviewed on receipt at site

Class	Lead time
L1	long lead ≥ 40 weeks
L2	26–39 weeks
L3	16–25 weeks
L4	< 16 weeks

5 · FAT and SAT outline of the DCS and SIS (PKG-12)

FAT at the vendor shop on the real cabinets with simulated inputs; SAT on site after installation, followed by the loop checks and the SIS function tests of DC3-V1-CM-PCC-001.

System	AI	AO	DI	DO	Total	Spare 20 % (per type, rounded up)	Installed
DCS	31	7	7	4	49	12	61
SIS	3	0	5	3	11	3	14

Instrument rows from tabs/einst/work/inl.json (DC3-V1-IN-INL-001); load list from tabs/eelec/work/loads.json (DC3-V1-EE-LL-001). Recommended additions not in this scope until approved: PT-221 (SIS AI), PSHH-221 (SIS —), XV-222 (SIS DO), ZSC-222 (SIS DI).

Step	FAT (vendor shop, simulated I/O)	SAT and site tests	Acceptance
1	Hardware, cabinets, documentation; power-up	Re-check after shipping; power from UPS-DCS / UPS-SIS	diagnostics clear
2	I/O count and tags against the INL (60 channels)	Loop checks: 42 loops	tag, range, unit as the INL
3	Redundancy, fail-over, diagnostics	Repeat with field I/O live	bumpless fail-over
4	9 DCS controllers simulated (direction, tuning, modes, fail-safe output)	Control valves stroked from the console	as the INL
5	23 alarm rows of the C&E	Alarm limits in the loop checks	limits as the alarm list
6	6 C&E cause rows with all effects and first-out	SIS function tests with real sensors and XVs	effects and delays as the C&E; XV stroke ≤ 2 s
7	DCS–SIS communication, sequence of events	Repeat on site	within one scan
8	Graphics against the P&ID	Operators trained on the OTS walk the graphics	every tag reachable
9	FAT punch list and report	SAT report, handed to commissioning	punch A closed

C&E function tests

Test	Cause	Setting / delay	Effects expected
CE-01	**HS-ESD** Emergency shutdown push-button (DCS / SIS screen)	0 s	XV-201 C, FIC-201 MAN, XV-202 C, XV-209 C, TIC-202 MAN, TIC-208 MAN, SIS TP, 1ST F, RED A, AUD A
CE-02	**FSLL-201** Feed flow low-low (FT-201): spare feed pump P-201B auto-start	15 m³/h · 5 s	P-201B ST, AMB A, AUD A
CE-03	**PSHH-221** Column pressure high-high trip on its own PT-221 (RECOMMENDED, not in the logic)	1.77 MPa g · 3 s	XV-202 (X), XV-209 (X), TIC-202 (X), TIC-208 (X), SIS (TP), 1ST (X), RED (X), AUD (X) — recommended, not in the logic (EC-07)

Test	Cause	Setting / delay	Effects expected
CE-04	**LSLL-202** C-201 sump level low-low trip (LS-202): heat off	5 % · 5 s	XV-202 C, XV-209 C, TIC-202 MAN, TIC-208 MAN, SIS TP, 1ST F, RED A, AUD A
CE-05	**LSHH-203** Reflux drum V-202 level high-high trip (LS-203): feed and heat off	92 % · 5 s	XV-201 C, FIC-201 MAN, XV-202 C, XV-209 C, TIC-202 MAN, TIC-208 MAN, SIS TP, 1ST F, RED A, AUD A
CE-06	**PDSL-230** Reflux pump differential pressure low (PDT-230): spare reflux pump P-202B auto-start	150 kPa · 3 s	P-202B ST, AMB A, AUD A
CE-07	**I-201** Open XV-209 / XV-202 refused: start permissive not met	—	SIS RS, AMB A

6 · Assumptions

- Project kick-off 04-Jan-27; mechanical completion of the last system week 78 (03-Jul-28); phase durations as in section 1.
- Lead times (purchase order to delivery on site, including shop work, FAT and transport), the default purchase-order weeks and the construction need offsets (weeks before MC) of section 2 are typical values for a single-column LPG unit, not quotations.
- Lead-time classes: L1 long lead ≥ 40 weeks; L2 26–39 weeks; L3 16–25 weeks; L4 < 16 weeks. A package is critical when its float to the need date is 4 weeks or less.
- Inspection levels 1–4 as in section 4 (company scheme, assumed). H = hold point, W = witness point, R = document review.
- Instrument-list rows are assigned to packages by their type, in the order control valves, SIS valves, PSV, field instruments, analyzers, piping specialities, DCS/SIS functions.
- I/O spare 20 %; XV stroke time ≤ 2 s (SRS assumption); SIS FAT scope = the C&E rows in the logic (PSHH-221 excluded, query EC-07).

Engineering queries raised

No.	Subject	Query	Proposed answer	Status
EC-01	P-201A/B run-in on water	Run on water, P-201 develops the same head (m) but 1.82 × the pressure and power of LPG (ρ 548.5 kg/m³): at 60 m³/h it absorbs 90.7 kW against a 75 kW motor, and its shut-off pressure on water is 4.22 MPa g against the 2.85 MPa g design of the casing, E-201 tubes and the 4"-P-202 / 6"-P-203 lines (hydrotest 3.705 MPa g).	No water run-in against the plant system. Run P-201A in on water only on a temporary loop with a full-flow relief set below 2.85 MPa g and the flow limited to 46 m³/h (motor), or commission the pumps directly on LPG after the solo run of the motor.	OPEN
EC-02	P-202A/B run-in on water	At 81 m³/h on water P-202 would absorb 43.9 kW (2.12 × the 20.7 kW on reflux at ρ 471.7 kg/m³) against a 30 kW motor; shut-off on water 1.51 MPa g is inside the 2.60 MPa g design.	Limit any water run to 47 m³/h by throttling the discharge, with the motor current watched; preferably commission on propane.	OPEN
EC-03	C-201 hydrotest weight for the foundation and skirt	Full of water C-201 weighs 76.6 t (empty 23.2 t + 53.4 m³ water) against about 27.6 t in operation (model holdup 4.4 t). A vertical field hydrotest also raises the bottom to 2.704 MPa g (water head 0.234 MPa over 23.82 m).	Design the foundation, skirt and anchor bolts for the hydrotest load case; test C-201 in the shop horizontally at 2.535 MPa g and in the field only after a repair, with civil confirmation.	OPEN
EC-04	Tightness test pressure	The N_2 tightness test is limited to 0.80 MPa g by the N_2 header (assumed); a test at the operating pressure (up to 1.740 MPa g in the column system) would also exceed 90 % of the PSV-201 set (1.62 MPa g) because the set is only 5.9 % above operating (EP-01).	N_2 test at the header pressure; the range above it is proven by a hydrocarbon leak check at each pressure step of the start-up (hot bolting rounds). Revisit when EP-01 (PSV set) is answered.	OPEN
EC-05	Long-lead items on the critical path	With the assumed lead times PKG-01 Column C-201 (44 weeks, float 2 weeks) and PKG-04 Pumps P-201A/B, P-202A/B (48 weeks, float 2 weeks) have ≤ 4 weeks of float to their construction need dates; one month of delay in their purchase orders delays mechanical completion.	Place the column and pump purchase orders on the FEED datasheets (DC3-V1-ME-DS-001) with a later technical freeze, expedite monthly, and plan the column erection as the first heavy lift.	OPEN
EC-06	Dryness criterion after the hydrotest	The design basis has no dryness criterion; −20 °C dew point is assumed. The feed arrives at −1 °C and propane auto-refrigerates to −42 °C on depressuring: free water would form ice / hydrates in valves and instrument taps.	Confirm the dew point (−20 °C or lower) with the propane product water specification; dry with dry N_2 after the hydrotest.	OPEN
EC-07	PSHH-221 in the SIS FAT scope	PSHH-221 is a recommended SIF (HAZOP, LOPA, SRS) but not in the logic; the SIS FAT tests 6 causes. Adding it after the FAT means a second FAT and re-validation.	Decide on PSHH-221 (PT-221, own nozzle) before the SIS configuration freeze so it is tested at the FAT.	OPEN
EC-08	Performance-test corrections and tolerances	The guarantees are stated at the design case (cooling water 330 m³/h at 15 °C, LP steam 0.45 MPa g, feed z = 0.5707); on the test day these differ. Tolerances assumed: capacity −1 %, steam +5 %.	Agree in the contract the correction curves (CW temperature, feed composition) and the measurement uncertainty; calibrate FI-201, FI-205, FI-206 and the analyzers before the test.	OPEN
EC-09	OTS start-up begins under fuel gas	The simulator's start-up procedure starts from the cold state under a fuel-gas blanket: N_2 purging, gassing-up and the oxygen check of SOP-C201-01 are not in the simulator.	Keep them as field steps in the commissioning procedure; brief the operators on the OTS for the steps after gassing-up only.	OPEN